Lab 4: Regression and Classification Evaluation Metrics

Part 1: Comprehensive Study of K-Nearest Neighbours (KNN) Classification using Breast Cancer Dataset and Comparison with Regression Evaluation Metrics

Aim

To implement KNN classification on the Breast Cancer dataset and analyze model performance using train-test split, heuristic K selection, cross-validation, ROC-AUC, and classification metrics. Also, to compare classification metrics with regression metrics studied in Linear Regression (Lab 3).

Dataset

Breast Cancer Wisconsin (Diagnostic) Dataset

From UCI : https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic

Or From kaggle :
https://www.kaggle.com/datasets/utkarshx27/breast-cancer-wisconsin-diagnostic-dataset

Or from csv :
https://drive.google.com/file/d/1poP5CatfiZGaRbGCAKPGe_wOBCGspDjA/view?usp=drive_link

- 569 samples
- 30 numerical features
- Target:
    - 0 → Malignant
    - 1 → Benign

Problem Statement

A healthcare analytics team is developing a predictive model for early cancer detection. You are required to build a KNN classifier, optimize its performance using different validation techniques, and compare classification evaluation metrics with regression evaluation metrics from Lab 3.

Tasks

Task 1: Data Preparation

1. Load the dataset using sklearn.

2. Convert into DataFrame and explore structure.
3. Check missing values and duplicates.
4. Apply feature scaling using StandardScaler and justify its importance.

Task 2: Train-Test Split Analysis

1. Split dataset into training and testing sets (80:20).
2. Repeat with 70:30 and 90:10 splits.
3. Compare performance variations across splits.
4. Analyze how dataset splitting affects model stability and generalization.

Task 3: KNN Model with Heuristic K Selection

3.1 Heuristic Method for K Selection

1. Compute initial K using heuristic rule ($K = \sqrt{n}$);where n = number of training samples
2. Use this value as a baseline K.

3.2 Model Training

1. Train KNN classifier using heuristic K value.
2. Experiment with nearby values of K ($K \pm 5$).
3. Plot accuracy vs K values.
4. Identify optimal K based on performance trend.

3. 3 Distance Matrix and Decision Boundary Mapping

1. Explain any two distance metrics used in KNN: **Euclidean Distance** and **Manhattan Distance**. Also mention when each distance metric is suitable.
2. Plot the **decision boundary** of the KNN classifier for different values of K, such as K = 1, K = 5, K = 10, and K = 20. Analyze how the decision boundary changes as K increase

Task 4: Cross Validation

1. Apply K-Fold Cross Validation (k = 5 or 10).
2. Compute mean accuracy for different K values.
3. Compare cross-validation results with train-test split results.
4. Select best K based on both heuristic + validation results.

Task 5: Classification Evaluation

Evaluate final model using:

- Accuracy

- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC Curve and AUC Score

Task 6: Comparative Study with Regression (Lab 3 Integration)

Reference: Recall **Lab 3 (Linear Regression)** where you evaluated:

- MAE
- MSE
- RMSE
- R^2 Score

Tasks:

1. Compare regression evaluation metrics with classification metrics.
2. Explain differences between:
    - Error-based evaluation (Regression)
    - Decision-based evaluation (Classification)
3. Compare:
    - R^2 Score vs Accuracy
    - RMSE vs F1 Score
    - MAE vs Confusion Matrix
4. Discuss differences in evaluation logic for:
    - Continuous prediction tasks
    - Classification tasks

Inference Requirement (short and precise -don't generate)

Write a detailed inference covering:

- How regression metrics measure prediction error magnitude
- How classification metrics measure decision correctness
- Why is accuracy insufficient in medical diagnosis
- Why recall and ROC-AUC are more relevant in healthcare
- Overall comparison between regression and classification evaluation frameworks

Task 7: Analytical Questions

1. Why is KNN called a lazy learning algorithm?
2. Why is feature scaling required in KNN?

3. Explain heuristic K selection using √n rule.
4. Why is cross-validation more reliable than a single train-test split?
5. How does K affect bias-variance trade-off?
6. Why is recall more important than accuracy in cancer prediction?
7. What is the limitation of very large K values?

Expected Outcome

- Understanding of KNN algorithm and distance-based learning
- Ability to tune K using heuristic + validation
- Knowledge of cross-validation
- Interpretation of ROC-AUC in medical classification
- Strong conceptual link between regression and classification evaluation metrics

Conclusion

Students should summarize:

- Optimal K value using √n and validation
- Effect of train-test split variations
- Model performance based on evaluation metrics
- Key differences between regression and classification evaluation
- Insights from Lab 3 vs current lab